

OCCURRENCE OF MONOGENEANS ON SELECTED CYPRINID FISHES FROM THE RIVER PENNA IN YSR DISTRICT, ANDHRA PRADESH

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ABSTRACT : The current study was the first widespread survey on the parasites of fishes from river Penna in YSR district of Andhra Pradesh. During the two years extensive survey (from July 2017 to June 2019), eight fish species of the cypriniformes family were examined. Total six dactylogyrid monogenean parasites were recorded which include *Dogielius catlaius* in *Cirrhinus ariza* and *Labeo catla*; *Dactylogyrus fotedari* in *Labeo calbasu* and *L. catla*; *Paradactylogyrus catalaius* in *L. rohita*; *Dactylogyrus lamellatus* in *L. dyocheilus*; *Dactylogyrus mrigala* in *Puntius sarana* and *Ancyrocephalus ghoshi* in *Salmostoma bacaila*. The prevalence of monogenean infestation was high in *C. ariza* (95%), *P. sarana* (82.3%) and *L. calbasu* (75%) and least in *S. bacaila* (30%). The overall prevalence of monogeneans in the investigated cyprinid fishes from Penna river was 70.6% and the mean intensity was 11.5. In addition, greatest intensity of infestation and pathological effect was with *D. fotedari* (17.6) in *L. calbasu*. To the best of our knowledge, this is the first report on monogenean parasite survey from cyprinid fishes of river Penna. All the parasite species represented the new geographical record for the parasite fauna of fishes in Kadapa, YSR district of Andhra Pradesh.

Key words : *Dogielius catlaius*, *Dactylogyrus fotedari*, *Paradactylogyrus catalaius*, *Dactylogyrus lamellatus*, *Dactylogyrus mrigala*, *Ancyrocephalus ghoshi*, cyprinid fish, Penna river.

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INTRODUCTION

Fishes are naturally infected with a variety of ectoparasites such as monogeneans, copepods, isopods and endoparasites such as digeneans, cestodes, acanthocephalans, and nematodes. Monogeneans are very crucial in fish pathology as they exhibit variable characteristics in hosting fishes, where they live as ectoparasites based on the physical and chemical characteristics of water. Monogeneans are considered as bioindicators and display health of the host and quality of the water. These monogeneans cause infestation, excessive mucus production and paves way for secondary bacterial infections in the host (Reed *et al*, 1996; Abid *et al*, 2016). There are two types of monogenoids infecting the freshwater fishes *i.e.* *Gyrodactylus* and *Dactylogyrus* species, which cause gyrodactylosis and dactylogyrosis diseases and result in mass mortality of fishes. There are only few studies on the monogenoids of cyprinids from various parts of India (Tripathi, 1959;

Agrawal, 1980; Dubey *et al*, 1997; Gupta *et al*, 2007; Pandey and Agarwal, 2008; Tak *et al*, 2014; Sujana and Shameem, 2015; Abid *et al*, 2016; Ramudu *et al*, 2018; Vijaysundardeva *et al*, 2018). However, less work was done on the fishes of river Penna (Indra, 2011) and their parasites (Modi and Vankara, 2018, 2019 and 2021; Vankara and Modi, 2021; Rafeeq *et al*, 2021). Penna is one of the non-perennial rivers flowing through Peninsular India with many tributaries such as Chitravathi, Kunderu, Papaghni, Sagileru and Cheyeru. It serves as a good habitat for a number of freshwater fishes, especially cyprinids. However, this river water is frequently polluted by domestic wastes from the neighboring settlements. In addition, the level of contamination is relatively high due to the cement factories and other industrial developments at that location which deteriorates the quality of the water. Hence, a timely survey on the parasite fauna of fish and effect of pollution on them is required to know much better about the parasite diversity.

MATERIALS AND METHODS

Fishes were procured from the sampling areas through gillnets and bagnets with the help of local fishermen for two years (from July, 2017 to June, 2019). In the current work, cyprinid fish specimens of four genera and eight species were examined (Table 1). Specifically, a total of 374 cyprinid fish specimens were examined for the presence of ectoparasites. All the monogenean species in each individual fish were identified and counted. Monogeneans were temporarily mounted using ammonium picrate-glycerin mixture and neutral red ($C_{15}H_{17}N_4Cl$), which helps in studying the sclerotized hard parts and the reproductive organs of the parasites (Malmberg, 1970). Parasite identification was carried out with the help of standard protocols (Gusev, 1976a, 1976b; Pandey and Agrawal, 2008). Lynx trinocular microscope (N-800M) was used to observe, identify and capture the microphotographs of temporarily stained parasites. The line diagrams were drawn with the aid of the attached drawing tube. Ocular micrometer measurements were depicted in micrometers (μm) using Image J software unless indicated (<https://imagej.nih.gov/ij/download.html>).

RESULTS AND DISCUSSION

In the present investigation, parasitological investigations of the gills of cyprinid species from river Penna of YSR district, Andhra Pradesh was carried out. A total of 6 dactylogyrid monogeneans were noticed which include *Dogielius catlaius* in *Cirrhinus ariza* and *Labeo catla* (Fig. 1), *Dactylogyrus fotedari* in *Labeo calbasu* and *Labeo catla* (Fig. 2), *Paradactylogyrus catalaius* in *Labeo rohita* (Fig. 3), *Dactylogyrus lamellatus* in *Labeo dyocheilus* (Fig. 4), *Dactylogyrus mrigala* in *Puntius sarana* (Fig. 5) and *Ancyrocephalus ghoshi* in *Salmostoma bacaila* (Fig. 6). All the observed monogeneans represented the initial geographical evidence for the parasite fauna of fishes in river Penna. A total of 374 specimens of 8 fish species were investigated, of which 264 (70.6%) were infected with monogeneans with a mean intensity of 11.5 (Table 1). Of the eight examined fish species, only *Cirrhinus cirrhosus* was uninfected while the remaining seven species were infected by at least one monogenean species.

Table 1 : Prevalence and intensity of infestation with monogeneans among cyprinid host fish species of River Penna, YSR Kadapa during the study period, July 2017- June 2019.

Name of the fish	Name of the parasite	No. of examined fish	No. of infected fish	No. of parasites	Prevalence (%)	Mean intensity	Mean abundance	Index of infection
<i>Cirrhinus cirrhosus</i> (Bloch, 1975)	-	15	0	0	0	0	0	0
<i>Cirrhinus ariza</i> (Buchmann, 1807)	<i>Dogielius catlaius</i> (Jain, 1961) Gusev, 1976	40	38	385	95	10.1	9.6	9.1
<i>Labeo calbasu</i> (Hamilton, 1822)	<i>Dactylogyrus fotedari</i> (Jain, 1960) Gusev, 1978	122	92	1623	75.4	17.6	13.3	10.0
<i>Labeo catla</i> (Hamilton, 1822)	<i>Dactylogyrus fotedari</i> (Jain, 1960) Gusev, 1978	55	40	401	72.7	10.0	7.3	5.3
<i>Labeo catla</i> (Hamilton, 1822)	<i>Dogielius catlaius</i> (Jain, 1961) Gusev, 1976	55	40	208	72.7	5.2	3.8	2.8
<i>Labeo rohita</i> (Hamilton, 1822)	<i>Paradactylogyrus catalaius</i> Thapar, 1948	57	39	256	68.4	6.6	4.5	3.1
<i>Labeo dyocheilus</i> (McClelland, 1839)	<i>Dactylogyrus lamellatus</i> Achmerow, 1952	25	16	25	64	1.6	1	0.6
<i>Puntius sarana</i> (Hamilton, 1822)	<i>Dactylogyrus mrigali</i> Gusev, 1976	40	33	128	82.5	3.9	3.2	2.6
<i>Salmostoma bacaila</i> (Hamilton, 1822)	<i>Ancyrocephalus ghoshi</i> Gusev, 1976	20	6	14	30	2.3	0.7	0.2
Total		374	264	3040				

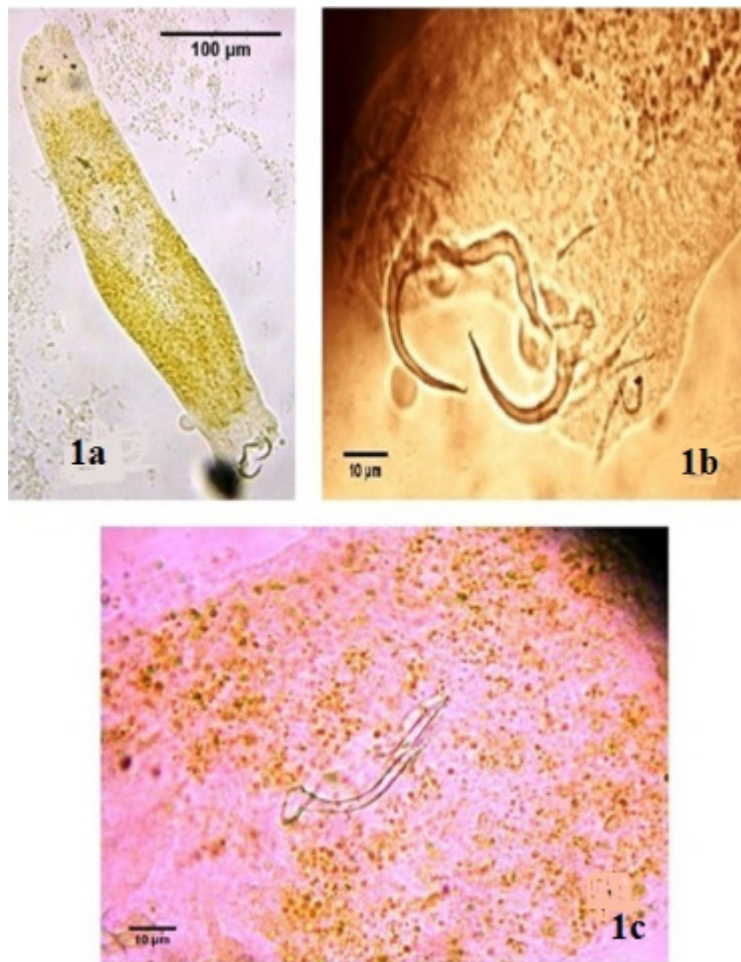


Fig. 1 : *Dogielius catlaius* (Jain, 1961) Gusev, 1976. **1a**: Microphotograph of *Dogielius catlaius* (Whole mount) 100X, **1b**: Microphotograph of opisthaptor 100X, **1c**: Microphotograph of copulatory complex 400X.

Cirrhinus ariza

40 specimens were examined, of which 38 (95%) fishes had infestation with *Dogielius catlaius* with a mean intensity of 10.13.

Dogielius catlaius (Jain, 1961) Gusev, 1976 (Fig. 1): Body elongated (280-700 × 60-150), five pairs of head organs, and two pairs of eye-spots. Pharynx rounded (40-70 × 50-70), intestine bifurcated and caeca confluent posteriorly. Peduncle short, stout and rounded. Haptor bilobed (70-100 × 40-130), with one pair of anchors, each with stout bifurcated base (20) triangular, pointed, superficial (50) and rounded deep roots (60), Shaft (20) and fine recurved point (20) in length. Transverse bar, stout (30-50 × 10), with a sharp bent in middle and rounded extremities. Each rounded extremity is 50-60. Each hook length is 20. Ovary round, inter-caecal and pro-testicular. Vagina dextral, highly sclerotized and have several slit-like opening. Testis ovoid, posterior and inter-caecal. Seminal vesicle found spindle-shaped. Copulatory complex stout and tubular copulatory tube with three finger-like process (40) and an accessory piece (20).

Prostate gland is in pear-shaped. Vitellaria densely spread from the pharynx up to the posterior end of the body.

Labeo calbasu

122 specimens were examined of which 92 (75.4%) fishes had infection with only one dactylogyrid monogenean i.e. *Dactylogyrus fotedari* with a mean intensity of 17.6.

Labeo catla

55 specimens were examined of which 40 (72.7%) fishes showed infestation with a mean intensity of 15.2. Two monogenean species were found which include *Dactylogyrus fotedari* in 35 fishes (63.6%) with mean intensity of 11.4 and *Dogielius catlaius* in 22 (40%) fishes with a mean intensity of 9.4.

Dactylogyrus fotedari (Jain, 1960) Gusev, 1978 (Fig. 2): Body of the worms small and stout to moderate measuring (380-580 × 70-100). Head organs two pairs with two pairs of eye-spots. Pharyngeal glands are on sides of pharynx. Intestine bifurcates and united posteriorly. Haptor wedge-shaped (50-100 × 70-90) with

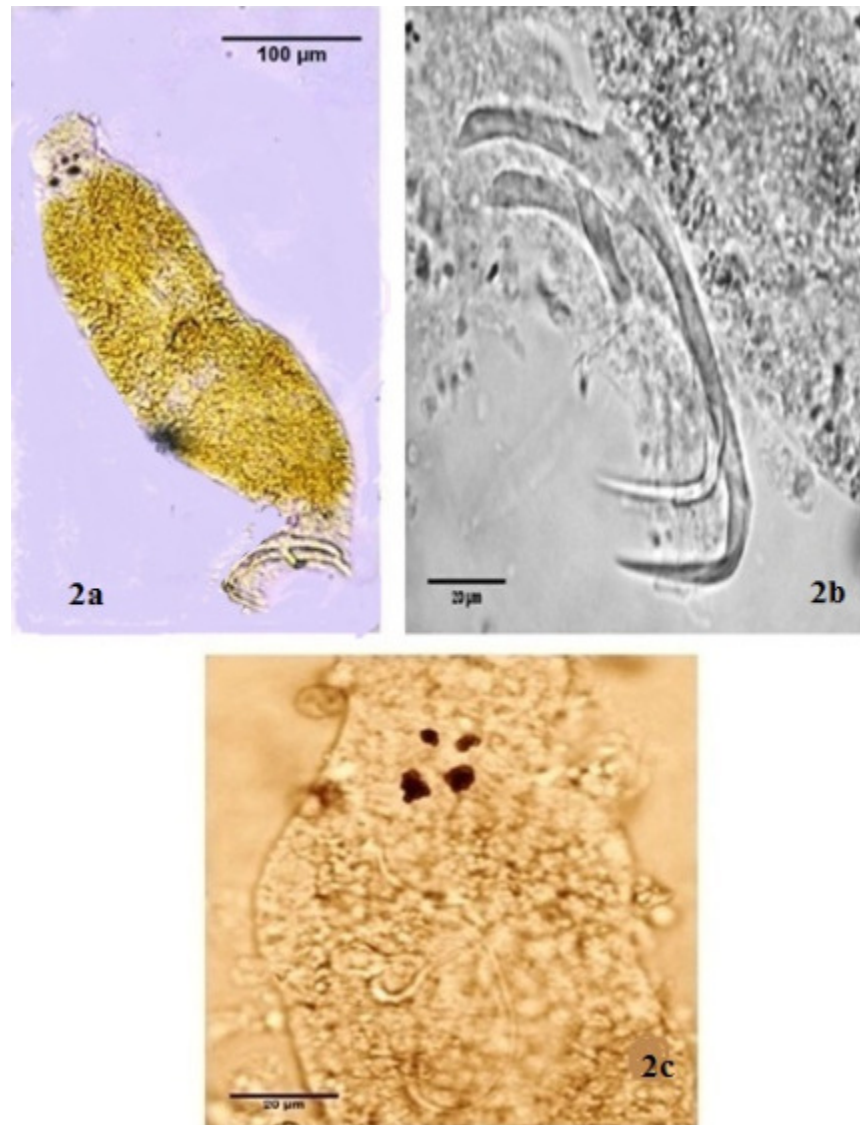


Fig. 2 : *Dactylogyrus fotedari* (Jain, 1960) Gusev, 1978. **2a:** Microphotograph of *Dactylogyrus fotedari* (Whole mount) 100X, **2b:** Microphotograph of opisthaptor 100X, **2c:** Microphotograph of copulatory complex 400X.

one pair of anchors. Anchors have long superficial root (40×10), insignificant deep root, strong shaft (30×10), and fine recurved point (20-30). Total length of median dorsal bar is $10-20 \times 10$. Six pairs of hooks arranged as in two sets. Total length and width of the hook is 10×10 and small wing (20-30). Anterior group of hooks are small with globular basic and thin shaft. Posterior group is large with an elongated base and small shaft. Ovary post-testicular, vagina sinistral, lightly sclerotized, opening into large seminal receptacle. Testis median and inter-caecal. Vas deferens narrow and S-shaped leading to pouch like seminal vesicle. Copulatory complex with a recurved, sclerotized copulatory tube (10-20) and a small parallel, pointed, articulated piece at base of accessory piece (20-10). Vitelline ducts join just anterior to seminal receptacle and total copulatory complex (20-40).

Labeo rohita

57 specimens were examined of which 39 (68.4%) fishes showed infestation *paradactylogyrus catalaius* with a mean intensity of 6.56.

Paradactylogyrus catalaius Thapar, 1948 (Fig. 3): The stout body of worm ($1200-1300 \times 190-200$). The anterior end bears cephalic lobes, three pairs of head organs and two pairs of eye-spots in the cephalic region. The prohaptor and haptor are set off by shallow constriction. Pharynx spherical (50×60). Opisthaptor is bilobed, well-demarcated from body ($140-150 \times 140-150$), with one pair of dorsal anchors (30×20), inner length (10-20), outer length (10), inner root (60-70), and outer root (10-20). Ventral anchors are absent and dorsal bar 'V' shaped (50×10). Onchium broad at upper end and pointed at lower end (40) and hooks seven pairs ($20 \times$

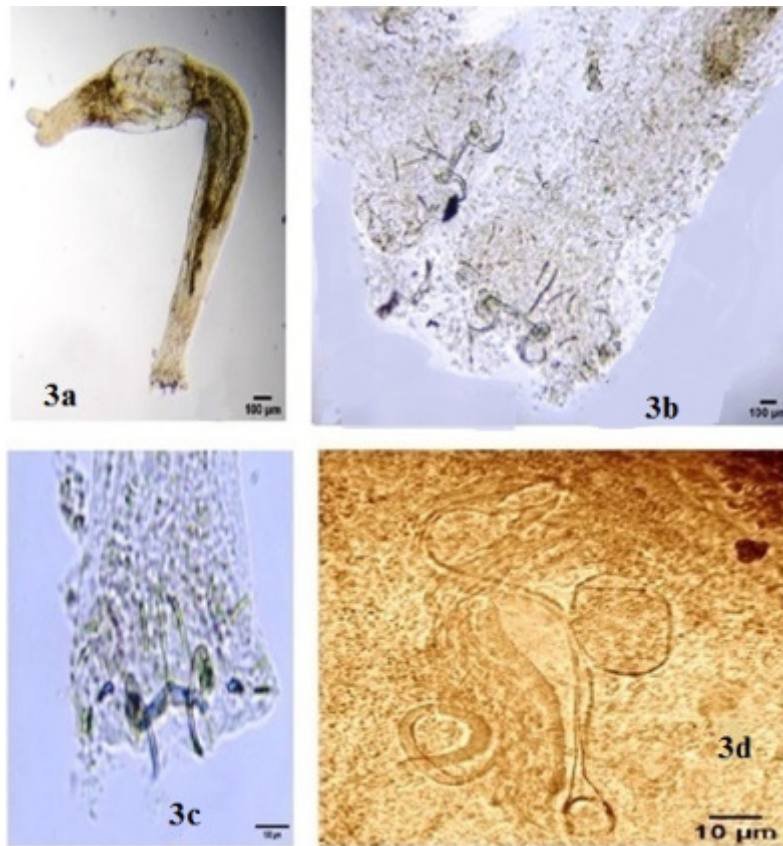


Fig. 3 : *Paradactylogyrus catalaius* Thapar, 1948. **3a:** Microphotograph of *Paradactylogyrus catalaius* (Whole mount) 100X, **3b:** Microphotograph of opisthaptor complex of parasites 100X, **3c:** Microphotograph of opisthaptor 400X; **3d:** Microphotograph of copulatory complex 400X.

10). Vaginal pore dextral, hook base (10), and shaft length (10), funnel-shaped and is connected by a long sclerotized vaginal tube to seminal receptacle. Ovary is mid-posterior and testis dorsal to the ovary. Vas-deferens arises from the end of the testis, runs anteriorly to loop around the left intestinal caecum forming finger-like blind appendage. Seminal vesicle opening at base of the copulatory tube by a long ejaculatory duct. Copulatory complex consists of a tubular, thick-walled copulatory tube (150) and a massive claw-like accessory piece (30). Single short prostatic reservoir opens at base of copulatory tube. Vitelline follicles dense, throughout the trunk and are absent in the region of reproductive organs.

Labeo dyocheilus

25 specimens were examined of which only 16 (64%) fishes were infected with *Dactylogyrus lamellatus* with a mean intensity of 1.56.

Dactylogyrus lamellatus Achmerow, 1952 (Fig. 4): Body (300-680 × 50-60). Pharynx (20-30 × 10-20), opisthaptor (50-80 × 10-30), anchors provided with joint pieces at the inner root (10-20), shaft length (30-40), inner root length (10-20), outer root length (30-40), base length (50-60), point recurved length (20), and dorsal bar (20 ×

10). Marginal hooks 7 pairs (10-20), testis (60 × 70), copulatory tube (20-30), accessory piece (20 × 30) and ovary round (10).

Puntius sarana

40 specimens were examined of which only 33 (82.5%) fishes were infected with *Dactylogyrus mrigali* with a mean intensity of 3.87.

Dactylogyrus mrigali, Gusev (1976) (Fig. 5): Body small (230-580 × 80-70). Haptor demarcated slightly from the body (80-10 × 60-80). Pharynx is 20 × 20. Intestinal caeca confluent and posterior to the testis. Anchors well-developed with roots and recurved points. Anchors base length (30) and shaft length (20), with curved point (10), inner root (20) and outer root (10). Dorsal bar has considerably curved backward medial part, saddle-shaped (20 × 10). Ventral bar five, ray-shaped (30-40 × 10). Hooks handle poorly demarcated from a pivot, projecting heel (20-30). Sixth Pair of hooks are shortest while the fourth pair hooks are the longest. Ovary posterior to the vaginal tube and testis posterior to the ovary. Vaginal armature dextral, curved tube (10-20 × 10). Vas deferens looped around the left intestinal limb. Copulatory complex (30), slightly curved slender tube (20) with accessory piece

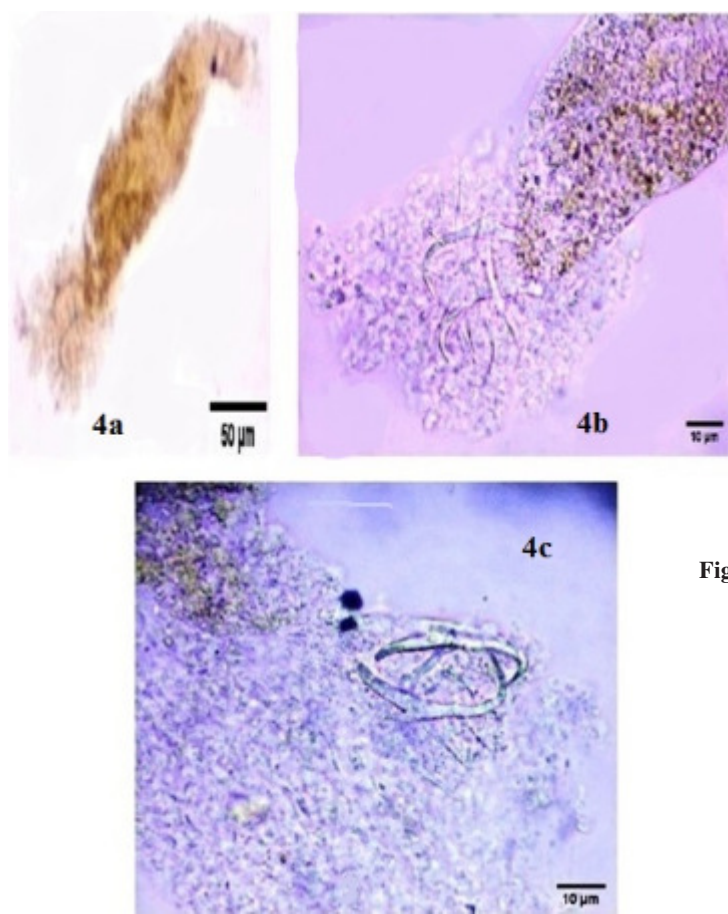
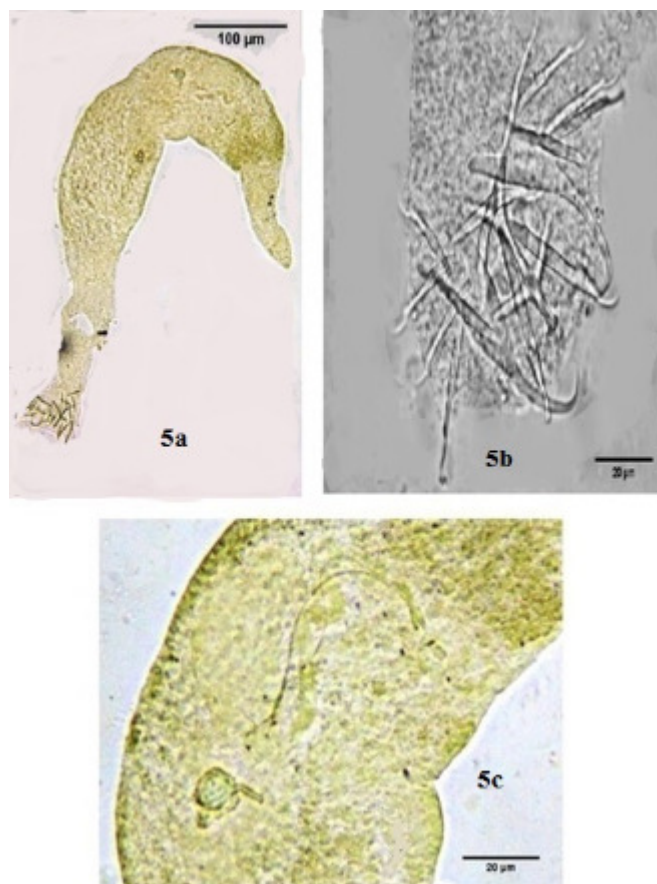


Fig. 4 : *Dactylogyrus lamellatus* Achmerow, 1952. **4a:** Microphotograph of *Dactylogyrus lamellatus* (Whole mount) 100X; **4b:** Microphotograph of opisthaptor 400X; **4c:** Microphotograph of copulatory complex 400X.

Fig. 5 : *Dactylogyrus mrigali* Gusev, 1976. **5a:** Microphotograph of *Dactylogyrus mrigali* (Whole mount) 100X; **5b:** Microphotograph of opisthaptor 400X; **5c:** Microphotograph of copulatory complex 400X.



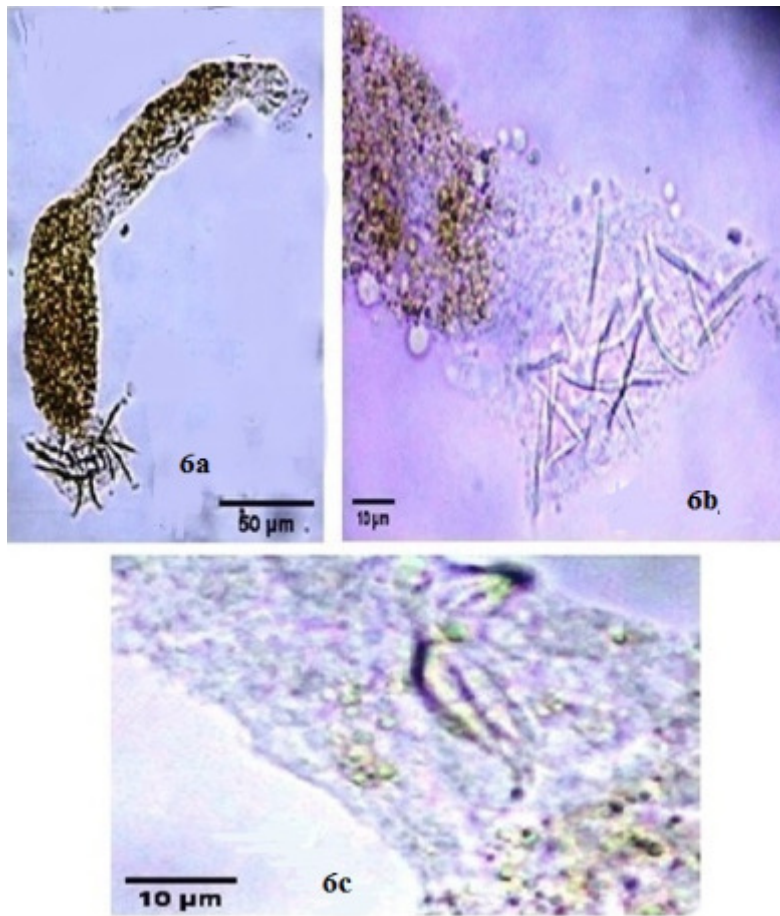


Fig. 6 : *Ancyrocephalus ghoshi* Gusev, 1976. **6a:** Microphotograph of *Ancyrocephalus ghoshi* (Whole mount) 100X, **6b:** Microphotograph of opisthaptor 400X, **6c:** Microphotograph of copulatory complex 400X.

narrowing towards end with bubble-like inflated initial part (10). Vitellaria extend from the pharynx to posterior end of the body.

Salmostoma bacaila

Only 20 specimens were examined of which only 6 specimens (30%) were infected with *Ancyrocephalus ghoshi* with a mean intensity of 2.3.

Ancyrocephalus ghoshi Gusev, 1976 (Fig. 6): Body small (240-310 × 40-50). The cephalic lobes distinct and head bear eye-spots. Pharynx spherical (20 × 20) and the opisthaptor measures, 40-50 × 50-80. The armature of opisthaptor (20 × 10) consists of two pairs of anchors i.e. dorsal and ventral. Each dorsal anchor consists of base (20), shaft (20), and distal curved point (10). Ventral anchor (20 × 10) arranged with inner root (10) and outer root (10). Ventral anchor basal part is 20 only. There are two connective bars that combine two pairs of dorsal and ventral anchors. Total length of the dorsal bar is 20 × 10 and ventral bar is 20 × 10. Each hooks measure 20 × 10 and differentiated into basal and shaft regions i.e. hook basal part (10-20) and shaft (10). Copulatory complex with tube (30-50), straight wide, goblet-shaped proximally,

and crescent-shaped distally, accessory piece cross-shaped.

Among the monogenean species on cyprinids, *Dogielius catlaius* showed the highest prevalence (95%) in *C. ariza* and greatest infestation was observed with *D. fotedari* (17.6) in *L. calbasu* and *L. catla* (11.4). The greatest number of parasite specimens was found in the cases of infection with *D. fotedari*.

DISCUSSION

Parasitism in fishes is considered to be a major constraint in aquaculture (Bandado-Reantaso *et al*, 2005; Palaq *et al*, 2017; Modi and Vankara, 2021). Disease outbreaks in aquaculture occur due to many factors such as host density, temperature, pH, dissolved oxygen, host size and sex which determine the host susceptibility to various parasitic infestations. Monogeneans have a privilege to transmit to other fishes due to their life-cycles by direct contact (Dogiel, 1961). Though, there is huge information on parasitic fauna of freshwater fishes around the world, still there is a big lacuna in the detailed investigations involving a specific group of fish, especially from the Rayalaseema region of the Southern India.

Hence, a thorough investigation on the parasite fauna of cyprinid fishes was carried out in the river Penna. Altogether eight species of cyprinid fishes were examined for the parasitological studies i.e., *Cirrhinus cirrhosus*, *Cirrhinus ariza*, *Labeo catla*, *Labeo calbasu*, *Labeo rohita*, *Labeo dyocheilus*, *Puntius sarana* and *Salmostoma bacaila*. The monogeneans are considered to be appropriate models to study the parasite diversification as they exhibit higher morphological and ecological diversity (Poulin, 2001). Dactylogyrids are highly diverse group of parasites whose biogeography and distribution is clearly linked to the evolutionary history of their cyprinid hosts (Simkova *et al*, 2004). Specifically, *Dactylogyrus* is the highly diverse group of the monogenoidea with more than 900 nominal species restricted to freshwater fishes of the cyprinidae family (Timofeeva *et al*, 1997). In the present investigation, six species of monogenoidea belonging to five genera were identified, three under the genus *Dactylogyrus*, one species each under the genus *Paradactylogyrus* and *Dogielius*, respectively. The species *Paradactylogyrus catalaius* Thapar, 1948 was synonymized as *Dactylogyrus catalaius* by Monaco and Mizelle (1955) and Gusev (1976a). However, Yamaguti (1961), Agrawal (1980) and Agrawal *et al* (2001) have retained a separate generic status for *Paradactylogyrus*. In the current work after a careful examination, it was felt to place the species under the separate genus *Paradactylogyrus*. In addition, *Dogielius catalaius* was found to be infecting both *Cirrhinus ariza* and *Labeo catla*. Of the three species of *Dactylogyrus* namely, *Dactylogyrus fotedari*, *Dactylogyrus lamellatus* and *D. mrigali*, *D. fotedari* showed high infestation rates in *Labeo calbasu* and *Labeo catla*. Only a rare infestation of *Ancyrocephalus ghoshi* was observed in *Salmostoma bacaila*. A thorough appraisal of the Indian monogeneoid literature disclosed the truth that there is an immense paucity of information on studies dealing with monogenean parasites infecting specific group of fishes. This kind of studies facilitate in knowing the co-evolutionary patterns and processes among host and parasites more precisely.

CONCLUSION

In conclusion, present study found six dactylogyrid monogenean parasitic species from cyprinid fishes of river Penna of YSR district, which were not previously examined parasitologically in this river. These parasites were considered to be the new records for Penna River. In the light of these data, the geographical distribution of these parasites is extended and such studies contribute to future research to protect and develop the ecological potential of Penna river.

Conflict of interest

The authors declare that they have no conflict of interest related to the work.

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